Properties of High dimensional Data.

Curse of dimensionality: to approximate function using interpolation, need a grid in $\mathbb{R}^n$, which grow exponentially with respect to d.

- to sample Grid with resolution ε, need point $(1/\varepsilon)^d$.

Sampling in high-dimensional unit ball.

close Ball in $\mathbb{R}^d$ $\mathbb{B}^d := \{\vec{x}\in\mathbb{R}^d : x_1^2+x_2^2+\dots+x_d^2 \le 1\}$

draw sample uniformally from a unit ball

Naive approach: Draw each coordinate x_i uniformally from $[-1,1]$, keep the sample if $\|\vec{x}\|_2^2 \le 1$.

However, not work. Why!

Acceptance prob: $\frac{Vol(\mathbb{B}^d)}{Vol([-1,1]^d)}$, $Vol(\mathbb{B}^d) = \frac{2\pi^{d/2}}{d\,\Gamma(d/2)} \xrightarrow{d\to\infty} 0$ exponentially

↳ converge to 0 even faster.

Correct way:

- Generate $X_1,\dots,X_d$ independently from $N(0,1)$ and let $\vec{X} = (X_1,\dots,X_d)^T\in\mathbb{R}^d$.
- let $Y = \frac{\vec{X}}{|\vec{X}|_2}$
- sample ρ with pdf dr^{d-1} ($\leftarrow r\in[0,1]$) independently from X
- $\vec{Z} = \rho\vec{Y}$ is then uniformally distributed in $\mathbb{B}^d$.

Remark: $X_i\sim N(0,1) \to f_{X_i}(x_i) = \frac{1}{\sqrt{2\pi}} e^{-x_i^2/2} \to f_{\vec{X}}(\vec{x}) = \frac{1}{(\sqrt{2\pi})^d} e^{-\Sigma x_i^2/2}$.

$\vec{X}$ is spherical symmetrical, $\to$ $Y = \frac{\vec{X}}{|\vec{X}|_2}$ distributed unif on S^{d-1} unit sphere;

The surface area change when radius different, need to 'scale' $\vec{Y}$ to different radius surface.

~~pdf expected~~ let $\vec{Z} = \vec{Y}\rho$.

suppose the Vol($\mathbb{B}^d$) is Ar^d for $\mathbb{B}^d$ with radius r,

$\Rightarrow$ expected pdf $\frac{1}{A}$ $\Rightarrow$ probability-finding around sphere at r.

$$\rho \quad \underbrace{Adr^{d-1}}_{\text{surface area}}\, dr = \text{marginal distribution in radial}$$

$$= f_\rho(r)\, dr \quad \Rightarrow dr^{d-1} \text{ is pdf of } \rho.$$

Concentration properties of Highdimensional Gaussian & Gaussian.

Annulus Theorem.

Gaussian Annulus Theorem.

let $X_i \overset{iid}{\sim} N(0,1)$, $\vec{X}=(x_1,\cdots x_d)^T$ is called a d-dimensional spherical Gaussian, denote as $N(0, I_d)$, I_d is ~~dxd~~ identical matrix $\in \mathbb{R}^{d\times d}$. For $\forall \beta \le \sqrt{d}$.

$$P(|\|X\|_2-\sqrt{d}| \ge \beta) \le 3e^{-\cancel{96}\cdot -c\beta^2}$$

Remarks: $E(\|X\|_2^2) = dE(X_i^2) = d \Rightarrow$ expected radius^2 at d.

this means $\vec{X}$ distributed concentrationally around $\sqrt{d}$.

Proof: $P(|\|X\|_2-\sqrt{d}| \ge \beta) = P((\|X\|_2-\sqrt{d}) \ge \beta) + P((\|X\|_2-\sqrt{d}) \le -\beta)$

note if $\|X\|_2-\sqrt{d} \ge \beta \Leftrightarrow \|X\|_2 \ge \beta+\sqrt{d}$, and $\|X\|_2^2-d=(\|X\|_2+\sqrt{d})(\|X\|_2-\sqrt{d})$

$\ge \beta(\|X\|_2+\sqrt{d}) \ge \beta\sqrt{d}$.

Similarly if $\|X\|_2-\sqrt{d} \le -\beta \Rightarrow \|X\|_2^2-d \le -\beta(\|X\|_2+\sqrt{d}) \le -\beta\sqrt{d}$.

$$\Rightarrow P(|\|X\|_2-\sqrt{d}| \ge \beta) \le P(\underbrace{|\|X\|_2^2-d| \overset{?}{\ge} \beta\sqrt{d}}_{\text{cover}}) \le 2\exp\left(-c\min\left(\beta\sqrt{d}, \frac{\beta^2 d}{d}\right)\right)$$

$$\le 3\exp(-c\beta^2).$$

the last step use tail bound for χ^2 and $\beta \le \sqrt{d} \Rightarrow \beta^2 \le \beta\sqrt{d}$.